MEDISAARTHI

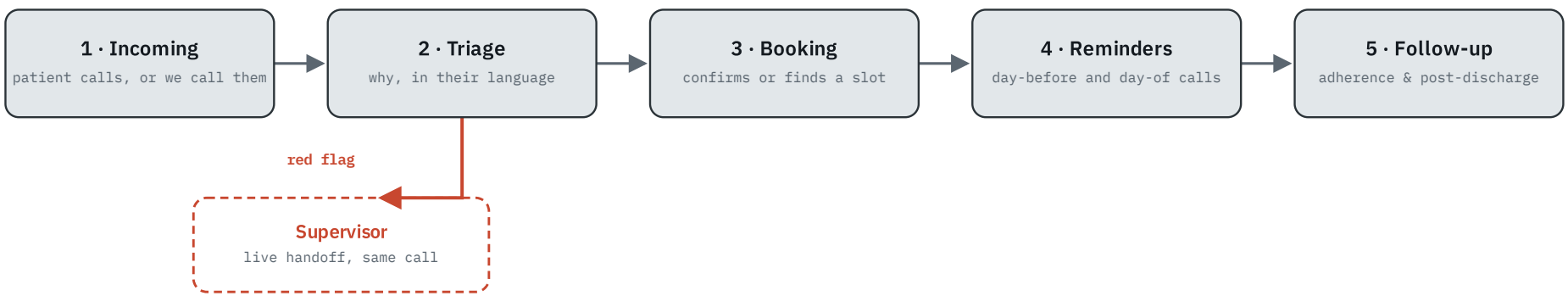
How it works

MediSaarthi is a phone-based AI assistant that talks to patients in their own language, handles the admin work hospitals struggle to staff for — triage questions, booking, reminders, follow-up calls — and hands off to a real person the instant anything sounds medically serious. It's built as the healthcare version of Novalantis's existing TalklyAI calling platform.

Some of this already works today. Some of it is still being built. This document walks through the whole thing once in plain terms, then again as an architecture diagram — and says clearly, throughout, which parts are which.

01 A patient's call, step by step

One conversation thread, from the first ring to the last follow-up call — with a human supervisor always reachable, not just notified afterward.



STEP 1

Incoming

The patient dials the hospital's number, or the hospital's schedule triggers an outbound call (a reminder, an adherence check). Either way, the system already knows which hospital and which language to use.

`reused — Talkly's existing call path`

STEP 2

Triage

A short conversation about why the patient is calling — never a diagnosis. Every sentence is checked against that hospital's own red-flag symptom list as it's said. A match hands the call to a person immediately, mid-conversation.

`new — the safety-critical piece`

STEP 3

Booking

Confirms, reschedules, or books a slot off the hospital's live calendar — while the patient is still on the line, so a cancelled slot can be offered to the next person right away.

`new — no calendar model exists yet`

STEP 4

Reminders

Day-before and day-of calls in the patient's language, with the ability to confirm, reschedule or cancel right there on the call instead of needing a second one.

`new scheduling, reused call-trigger pattern`

STEP 5

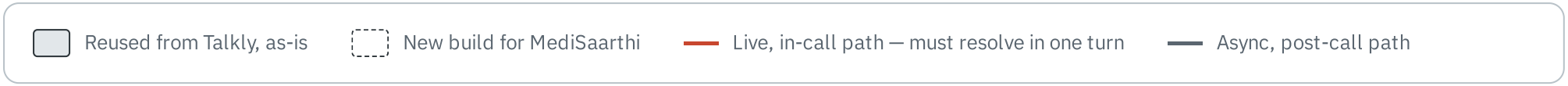
Follow-up

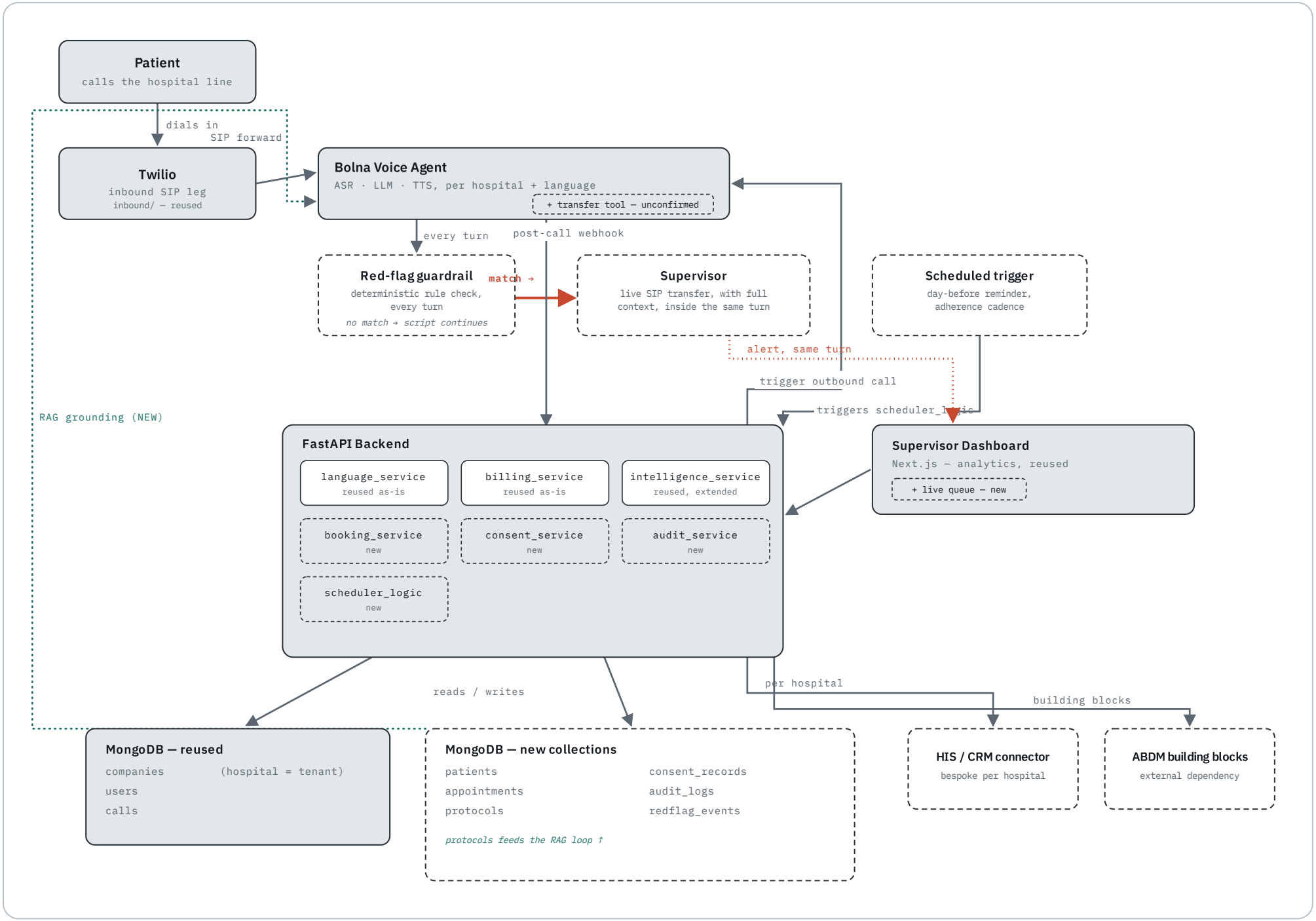
Scheduled medication reminders and post-discharge check-ins on a recurring cadence, so a missed response shows up for a supervisor instead of silently disappearing.

new — today's version only drafts emails

02 The full picture

Same flow, drawn as the system that runs it. Every box either exists in Talkly's codebase today or doesn't — the border style says which. The red path is the one that has to finish inside a single conversation turn; everything else can afford to be a webhook that runs after the call ends.




Patient
calls the hospital line
dials in
SIP forward
Twilio
inbound SIP leg
inbound/ — reused
Bolna Voice Agent
ASR · LLM · TTS, per hospital + language
+ transfer tool — unconfirmed
every turn
post-call webhook
Red-flag guardrail
match →
deterministic rule check, every turn
no match → script continues
Supervisor
live SIP transfer, with full context, inside the same turn
Scheduled trigger
day-before reminder, adherence cadence
alert, same turn
trigger outbound call
triggers scheduler_logic
RAG grounding (NEW)
FastAPI Backend
language_service
reused as-is
billing_service
reused as-is
intelligence_service
reused, extended
booking_service
new
consent_service
new
audit_service
new
scheduler_logic
new
Supervisor Dashboard
Next.js — analytics, reused
+ live queue — new
reads / writes
per hospital
building blocks
MongoDB — reused
companies (hospital = tenant)
users
calls
MongoDB — new collections
patients
appointments
protocols
consent_records
audit_logs
redflag_events
protocols feeds the RAG loop ↑
HIS / CRM connector
bespoke per hospital
ABDM building blocks
external dependency

Solid, filled boxes exist in Talkly's codebase today and are reused with little or no change.
Dashed, outline boxes are net-new for MediSaarthi.